SK015

Chemistry 1
Semester I
Session 2026/2027
2 hours

Kimia 1
Semester I
Sesi 2026/2027
2 jam

CHEMISTRY UNIT
LABUAN MATRICULATION COLLEGE
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KIMIA 1

2 Jam

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DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO

INSTRUCTIONS TO CANDIDATE:

This question paper consists of **6** questions. Answer **all** questions.

All steps must be shown clearly. Use a new page for each question.

Maximum marks awarded are shown in brackets at the end of each question or section.

The use of an electronic calculator is permitted.

You are allowed to cry but please do it quietly.

Regretting not studying won't change anything. Please remember this feeling the next time you want to skip your study session and binge on your K-Drama or push your rank.

I wish you ALL THE BEST and GOOD LUCK

QUESTION	MARKS
1	/21
2	/10
3	/17
4	/09
5	/09
6	/14
TOTAL	**/80**

TABLE OF RELATIVE ATOMIC MASSES

Element	Symbol	Proton number	Relative atomic mass
Aluminium	Al	13	27.0
Silver	Ag	47	107.9
Argon	Ar	18	40.0
Arsenic	As	33	74.9
Gold	Au	79	197.0
Barium	Ba	56	137.3
Beryllium	Be	4	9.0
Bismuth	Bi	83	209.0
Boron	B	5	10.8
Bromine	Br	35	79.9
Iron	Fe	26	55.9
Fluorine	F	9	19.0
Phosphorus	P	15	31.0
Helium	He	2	4.0
Mercury	Hg	80	200.6
Hydrogen	H	1	1.0
Iodine	I	53	126.9
Cadmium	Cd	48	112.4
Potassium	K	19	39.1
Calcium	Ca	20	40.1
Carbon	C	6	12.0
Chlorine	Cl	17	35.5
Cobalt	Co	27	58.9
Krypton	Kr	36	83.8
Chromium	Cr	24	52.0
Copper	Cu	29	63.6
Lithium	Li	3	6.9
Magnesium	Mg	12	24.3
Manganese	Mn	25	54.9
Sodium	Na	11	23.0
Neon	Ne	10	20.2
Nickel	Ni	28	58.7
Nitrogen	N	7	14.0
Oxygen	O	8	16.0
Platinum	Pt	78	195.1
Lead	Pb	82	207.2
Protactinium	Pa	91	231.0
Radium	Ra	88	226.0
Radon	Rn	86	222.0
Rubidium	Rb	37	85.5
Selenium	Se	34	79.0
Cerium	Ce	58	140.1
Caesium	Cs	55	132.9
Silicon	Si	14	28.1
Scandium	Sc	21	45.0
Tin	Sn	50	118.7
Antimony	Sb	51	121.8
Strontium	Sr	38	87.6
Sulphur	S	16	32.1
Uranium	U	92	238.0
Tungsten	W	74	183.9
Zinc	Zn	30	65.4

LIST OF SELECTED CONSTANT VALUES

Ionisation constant for water at 25°C	K_w	$= 1.00 \times 10^{-14}$ mol^2 dm^{-6}
Molar volume of gases	V_m	$= 22.4$ dm^3 mol^{-1} at STP $= 24$ dm^3 mol^{-1} at room temperature
Speed of light in a vacuum	c	$= 3.0 \times 10^8$ m s^{-1}
Specific heat of water		$= 4.18$ kJ kg^{-1} K^{-1} $= 4.18$ J g^{-1} K^{-1} $= 4.18$ J g^{-1} °C^{-1}
Avogadro's number	N_A	$= 6.02 \times 10^{23}$ mol^{-1}
Faraday constant	F	$= 9.65 \times 10^4$ C mol^{-1}
Planck constant	*h*	$= 6.6256 \times 10^{-34}$ J s
Rydberg constant	R_H	$= 1.097 \times 10^7$ m^{-1} $= 2.18 \times 10^{-18}$ J
Molar of gases constant	R	$= 8.314$ J K^{-1} mol^{-1} $= 0.08206$ L atm mol^{-1} K^{-1}
Density of water at 25°C	ρ	$= 1$ g cm^{-3}
Freezing point of water		$= 0.00$ °C
Vapour pressure of water at 25°C	P_{water}	$= 23.8$ torr

UNIT AND CONVERSION FACTOR

VOLUME	1 liter = 1 dm^3 1 mL = 1 cm^3
ENERGY	1 J = 1 kg m^2 s^{-2} = 1 N m = 1×10^7 erg 1 calorie = 4.184 J 1 eV = 1.602×10^{-19} J
PRESSURE	1 atm = 760 mmHg = 760 torr = 101.325 kPa = 101 325 N m^{-2}
OTHERS	1 faraday (F) = 96 500 coulomb 1 newton (N) = 1 kg m s^{-2}

Answer **ALL** questions.

1 (a) A naturally occurring sample of magnesium, Mg is analysed by using mass spectrometer and found to contain 78.6 % ^{24}Mg, 10.1 % ^{25}Mg and 11.3 % ^{26}Mg. Calculate the relative atomic mass of magnesium in this sample.

[2 *marks*]

(b) An organic compound ***A*** contains of carbon, hydrogen and chlorine only. Complete combustion of a 2.54 g sample of compound ***A*** produces 3.52 g of carbon dioxide CO_2 and 1.44 g of water H_2O. Given that the molar mass of compound ***A*** is 127.0 g mol^{-1}, determine its empirical formula and molecular formula of ***A***.

[6 *marks*]

(c) A concentrated laboratory stock solution of nitric acid, HNO_3, has a concentration of 68.0% by mass and a density of 1.41 g cm^{-3}. Calculate the molarity of this stock solution.

[4 *marks*]

(d) The following redox reaction occurs in an acidic medium:

$$CrO_4^{2-} + SO_3^{2-} \rightarrow Cr(OH)_3 + SO_4^{2-}$$

Write a balanced equation for the above reaction.

[3 *marks*]

(e) Silicon metal is extracted industrially from silicon dioxide by reacting it with magnesium metal at high temperature according to the following equation:

$$SiO_2\ (s) + 2\ Mg\ (s) \rightarrow Si\ (s) + 2\ MgO\ (s)$$

In an experiment, 30 g of silicon dioxide is reacted with 28 g of magnesium metal. If 10.5 g of silicon metal is collected from the reaction, calculate the percentage yield of silicon.

[6 *marks*]

2 (a) A line in the emission spectrum of hydrogen occurs at a wavelength of 1094 nm. This spectral line is produced when an electron drops from n=6 to a lower energy level.

(i) Determine the transition of the electron that forms this line.

(ii) Calculate the energy of the electron at the excited state.

(iii) State the name of the spectral series in which the line occurs.

[6 *marks*]

(b) A cation $\boldsymbol{D^{2+}}$ contains 20 electrons.

(i) Write the electronic configuration for element ***D***.

(ii) Draw the shape of orbital occupied by the highest energy electron in atom ***D***.

(iii) Determine the number of unpaired electron(s) present in atom ***D***.

(iv) Determine the total number of electrons in atom ***D*** that occupy orbitals with azimuthal quantum number ℓ=1.

[4 *marks*]

3 (a) Nitrogen forms various oxides and oxoanions including nitrogen dioxide (NO_2) and nitrite ion (NO_2^-).

(i) Draw the Lewis structures for nitrogen dioxide, NO_2 and nitrite ion, NO_2^-.

(ii) State which of the species in (i) does not obey the octet rule. Explain.

[4 *marks*]

(b) Iodine pentafluoride, IF_5 was once tested as a rocket propellent igniter.

(i) Based on Valence Shell Electron Pair Repulsion (VSEPR) theory, predict the molecular shape of IF_5.

(ii) Deduce whether a molecule of IF_5 is polar or non-polar.

[5 *marks*]

(c) Methanal, CH_2O and methanol, CH_3OH are both covalent compounds containing carbon, hydrogen and oxygen. The carbon-oxygen bond in methanal is shorter than methanol.

(i) Draw the Lewis structure of methanal, CH_2O and state the type of hybridisation of the central atom.

(ii) Draw the overlapping orbitals of methanal, CH_2O.

(iii) Methanal boils at -19 ^oC but methanol boils at 64.7 ^oC. Explain.

[8 *marks*]

4 (a) Ammonium nitrous, NH_4NO_2 decomposes upon heating to form nitrogen gas and water.

$$NH_4NO_2\,(s) \rightarrow N_2\,(g) + 2\,H_2O\,(l)$$

When a sample of NH_4NO_2 was heated and decomposed, 0.522 L of nitrogen gas was collected over water at 26 °C and 755 torr. If the vapour pressure of water is 25 torr at 26 °C, calculate the mass of nitrogen gas collected.

[5 *marks*]

(b) Four liquids, ***E***, ***F***, ***G*** and ***H***, have approximately similar molar masses. The intermolecular forces present in each liquid are shown in the table below.

Liquid	**Intermolecular Forces**		
	London Dispersion Forces	Dipole-Dipole Interaction	Hydrogen Bond
E	√		
F	√	√	
G	√	√	√
H	√	√	

Identify which liquid has:

(i) the highest vapour pressure.

(ii) the highest viscosity.

[2 *marks*]

(c) The solid-liquid line in the phase diagram of water exhibits an anomalous negative slope. Explain.

[2 *marks*]

5 Antimony (III) sulphide, Sb_2S_3, reacts with hydrogen gas at high temperatures according to the following equation:

$$Sb_2S_3\,(s) + 3H_2\,(g) \rightleftharpoons 2Sb\,(s) + 3H_2S\,(g) \qquad \Delta H > 0$$

A 2.50 L container containing 0.01 mol of solid antimony (III) sulphide and 0.01 mol of hydrogen gas is allowed to reach equilibrium at a constant temperature of 440 °C. At equilibrium, 4.3×10^{-3} mol of H_2S is found in the container.

(a) Calculate the value of equilibrium constant, K_c and K_p at 440 °C.

[7 marks]

(b) Suggest two changes to the experimental conditions that can be made to increase the production of H_2S gas.

[2 marks]

6 (a) Pyridine, C_5H_5N, is a weak base. A 0.10 M aqueous solution of pyridine has a pH of 9.10 at 25 °C. Determine the base dissociation constant, K_b of pyridine.

[5 *marks*]

(b) In an experiment, 12.0 mL of 0.10 M HCl solution is added to 20.0 mL of 0.10 M aniline, $C_6H_5NH_2$ solution. Calculate the pH of the resulting solution. [base dissociation constant, K_b of $C_6H_5NH_2 = 4.3\times10^{-10}$]

[6 *marks*]

(c) A solution contains 0.001 M Mg^{2+} and 0.001 M OH^-. Predict whether a precipitate of $Mg(OH)_2$ will form. [Given solubility product, K_{sp} of $Mg(OH)_2 = 1.8\times10^{-11}$]

[3 *marks*]

END OF QUESTION PAPER